WORKSHEET

Differentiating trigonometric functions

1 Differentiate the following trigonometric functions.

a $\sin(3x)$

b $\tan(7x)$

c $\cos(x)$

d $3\tan(x^4)$

e $\sin(x^2 + 5x - 1)$

f $5\cos(6x^3 - 4)$

g $\sin(x)$

h $4\tan\left(\frac{1}{2}x\right)$

i $-\cos(2x^6 + x^3 - 5x)$

j $\tan[\sin(x)]$

k $2\sin\left(\frac{1}{3}x\right)$

l $\tan(x)$

m $\sin(4x^3)$

© Cengage Learning Australia Pty Ltd 2014

n $3\cos(7-2x)$

o $5\tan(x)+\sin(x)$

p $2\cos(x)-\frac{1}{3}\tan(6x)$

q $\frac{1}{4}\tan(8x^5)$

r $7\sin[\cos(x)]+\cos(7x^3-4)$

s $\tan(x^4)-\sin(5x^2)$

t $-5\sin[\tan(x)]$

u $-\frac{3}{2}\cos\left(\frac{1}{3}x^2\right)+3\tan(x^3)$

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** $3\cos(3x)$

b $\dfrac{7}{\cos^2(7x)}$

c $-\sin(x)$

d $\dfrac{12x^3}{\cos^2\left(x^4\right)}$

e $(2x+5)\cos(x^2+5x-1)$

f $-90x^2\sin(6x^3-4)$

g $\cos(x)$

h $\dfrac{2}{\cos^2\left(\frac{1}{2}x\right)}$

i $(12x^5+3x^2-5)\sin(2x^6+x^3-5x)$

j $\dfrac{\cos(x)}{\cos^2[\sin(x)]}$

k $\frac{2}{3}\cos\left(\frac{1}{3}x\right)$

l $\dfrac{1}{\cos^2(x)}$

m $12x^2\cos(4x^3)$

n $6\sin(7-2x)$

o $\dfrac{5}{\cos^2(x)}+\cos(x)$

p $-2\sin(x)-\dfrac{2}{\cos^2(6x)}$

q $\dfrac{10x^4}{\cos^2\left(8x^5\right)}$

r $-7\sin x\cos[\cos(x)]-21x^2\sin(7x^3-4)$

s $\dfrac{4x^3}{\cos^2\left(x^4\right)}-10x\cos(5x^2)$

t $\dfrac{-5}{\cos^2(x)}\cos[\tan(x)]$

u $x\sin\left(\frac{1}{3}x^2\right)+\dfrac{9x^2}{\cos^2\left(x^3\right)}$

© Cengage Learning Australia Pty Ltd 2014